

Early Identification of Heart Attack vs Gastritis Using AI/ML

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Project Proposal Report

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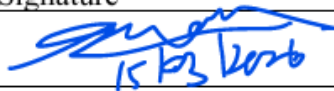
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgement is given in the text.

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ABSTRACT

Chest pain is one of the most common symptoms reported in emergency departments and primary healthcare settings, yet differentiating between cardiac-related chest pain and non-cardiac causes such as gastric disorders remains a significant clinical challenge, particularly during early diagnosis. Gastric conditions, including gastritis, gastroesophageal reflux disease (GERD), and peptic ulcer disease, can produce symptoms that closely resemble heart attack symptoms, which may lead to misdiagnosis, delayed treatment, and unnecessary medical procedures. Therefore, an intelligent decision support system capable of accurately distinguishing between cardiac and gastric causes of chest pain is highly valuable. Although recent studies have applied artificial intelligence and machine learning for cardiovascular disease detection and medical image analysis, most existing systems focus on either cardiac or gastrointestinal diseases independently, and only a limited number of approaches integrate multiple diagnostic data sources, such as patient symptoms, cardiac signals, and medical imaging, within a single framework. This research addresses this gap by proposing an AI-assisted diagnostic system that integrates multiple clinical inputs to support accurate differentiation between cardiac and gastric chest pain. The proposed system consists of four main components: patient questionnaire analysis, cardiac signal analysis, coronary blockage detection, and gastric disease analysis. Component 4, the Gastric Analysis and Decision Support Engine, uses symptom triggers and gastroscopy images to identify gastric conditions using deep learning techniques, specifically the EfficientNet-B4 architecture for classifying gastric diseases from endoscopy images. Image preprocessing is performed using OpenCV and data augmentation techniques, while Grad-CAM is applied to generate visual explanations for model predictions. In addition, a rule-based recommendation module generates treatment suggestions based on the detected disease and severity level. The expected outcome of this research is an AI-driven clinical decision support system that improves diagnostic accuracy and assists medical students in understanding the clinical reasoning process for chest pain diagnosis. Socially, the system may help reduce misdiagnosis and improve patient management, while commercially it has potential applications in healthcare decision-support platforms and medical education technologies.

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
CAD	Coronary Artery Disease
CVD	Cardiovascular Disease
ECG	Electrocardiogram
LAD	Left Anterior Descending Artery
RCA	Right Coronary Artery
LCX	Left Circumflex Artery
LMCA	Left Main Coronary Artery
PCI	Percutaneous Coronary Intervention
CABG	Coronary Artery Bypass Graft
IoU	Intersection over Union
DSC	Dice Similarity Coefficient
MAE	Mean Absolute Error
DSR	Design Science Research
REST	Representational State Transfer
SDG	Sustainable Development Goals
GERD	Gastroesophageal Reflux Disease
CNN	Convolutional Neural Network

Table 1:LIST OF ABBREVIATIONS

LIST OF APPENDICES

Appendix Title	Appendix Title
Appendix A	Survey Instruments
Appendix B	Preliminary UI mockups
Appendix C	Risk Analysis
Appendix D	Ethical Documents

Table 2:LIST OF APPENDICES

1. INTRODUCTION

1.1. Background and Context

Cardiovascular diseases (CVDs) remain one of the leading causes of mortality worldwide, with coronary artery disease (CAD) being a major contributor to heart attacks and other life-threatening cardiac conditions. According to global health statistics, millions of people die each year due to complications caused by blocked coronary arteries, highlighting the importance of timely diagnosis and treatment. In many clinical situations, patients who experience chest pain may initially assume their symptoms are related to gastric problems rather than cardiac conditions, which can delay proper diagnosis and increase the risk of severe complications. Therefore, early detection of coronary artery blockages and accurate treatment planning are essential for improving patient outcomes.

Coronary angiography is considered the gold standard diagnostic method for identifying coronary artery blockages. This imaging technique allows cardiologists to visualize blood flow within coronary arteries and determine the location and severity of arterial narrowing. However, interpreting angiogram images requires specialised knowledge and experience because the images are complex and difficult to analyse for non-experts. In many healthcare environments, particularly in developing regions, the limited availability of experienced cardiologists may lead to delays or inconsistencies in diagnosis. Additionally, traditional angiogram interpretation relies heavily on manual visual inspection, which can be time-consuming and subject to human variability.

Recent advancements in artificial intelligence (AI), machine learning, and computer vision have shown strong potential for improving medical image analysis and supporting clinical decision-making. Deep learning models such as Convolutional Neural Networks (CNNs) have demonstrated high accuracy in analysing medical images and detecting abnormalities. Several studies have applied AI techniques to coronary angiography analysis to automatically quantify coronary stenosis and identify arterial abnormalities with high precision [1], [2]. These technologies can assist clinicians by automatically identifying pathological patterns in medical imaging data, thereby improving diagnostic efficiency and reducing interpretation errors.

However, many existing AI systems primarily focus on detecting the presence of cardiovascular disease and often do not provide detailed clinical insights such as the affected coronary artery, blockage severity, or appropriate treatment options [3], [4]. To address these limitations, this research proposes an AI-assisted Angiogram Evaluation and Treatment Recommendation component as part of a broader intelligent diagnostic workflow designed to differentiate cardiac and gastric causes of chest pain.

The overall system integrates several AI-based modules that analyse patient symptoms, diagnostic tests, and imaging data to support clinical decision-making. The workflow begins with patient screening and risk assessment, followed by ECG and troponin validation, cardiac functional testing, and angiogram analysis for confirmed cardiac cases. This multi-stage diagnostic workflow supports clinical decision-making and ensures that patients follow the correct diagnostic pathway before treatment decisions are made.

Within this workflow, Component 3 focuses on automated analysis of coronary angiogram images and reports, where the system identifies affected coronary arteries, estimates the percentage of blockage (stenosis), and classifies disease severity. Based on this analysis, the system recommends appropriate treatment options such as medical management, percutaneous coronary intervention (PCI), or coronary artery bypass graft surgery (CABG).

In addition, the system generates visual explanations by highlighting blockage regions in angiogram images and creating simplified 3D visualisations of coronary arteries to improve understanding for both clinicians and patients. This approach addresses limitations of traditional angiogram interpretation, which relies heavily on specialist expertise and lacks automated tools for quantifying blockage severity or explaining results visually.

This research contributes to United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-Being, which focuses on reducing premature deaths from non-communicable diseases and improving access to advanced healthcare technologies. By enabling faster detection of coronary artery blockages and supporting treatment decision-making, the proposed system has the potential to enhance clinical efficiency and improve patient care.

From a research perspective, this study aligns with the Artificial Intelligence and Data Science research cluster, particularly within the fields of medical image processing, deep learning, and clinical decision support systems. The research investigates how computer vision and machine learning techniques can be applied to angiogram image analysis to detect coronary artery stenosis and generate explainable diagnostic outputs, contributing to the advancement of AI-assisted healthcare systems.

1.2. Literature Review

Recent advancements in artificial intelligence (AI) have significantly improved healthcare diagnostics by enabling automated analysis of complex medical data. In the context of chest pain diagnosis, several studies have proposed AI-based diagnostic frameworks that integrate patient symptoms, medical tests, and imaging data to support clinical decision-making. These systems analyse patient questionnaires, ECG signals, biomarkers, and medical images using machine learning and deep learning models to distinguish cardiac-related chest pain from non-cardiac causes. Such integrated diagnostic approaches demonstrate how AI can combine multiple data sources to improve diagnostic accuracy and enhance clinical workflow efficiency.

A considerable amount of research has focused on applying deep learning techniques to analyse coronary angiography images for detecting coronary artery disease. Convolutional Neural Networks (CNNs) are widely used to identify patterns in angiogram images and automatically detect arterial narrowing or stenosis. For example, automated coronary angiography interpretation systems such as CathAI have demonstrated the ability to detect coronary stenosis with high accuracy using deep learning techniques [2]. Similarly, research studies such as DeepCoro have applied AI-based video analysis models to evaluate coronary artery stenosis from angiographic imaging data [3].

Another important research direction involves coronary artery segmentation and stenosis quantification. Computer vision techniques are used to extract coronary artery structures from angiographic images and analyse vessel diameter to detect stenotic regions. Segmentation methods such as U-Net help identify major coronary arteries including the Left Anterior Descending (LAD), Right Coronary Artery (RCA), and Left Circumflex (LCX). Automated image-based stenosis estimation techniques have been proposed to quantify coronary artery narrowing using deep learning and end-to-end learning models [4], [5].

In addition to angiogram-based analysis, other studies have explored the use of CT angiography and advanced imaging methods for detecting coronary plaque and predicting cardiac risk. Deep learning-enabled coronary CT angiography models have demonstrated the ability to quantify plaque and stenosis severity while also predicting future cardiac risk [6]. Similarly, automated AI-based diagnostic systems have been developed to detect coronary artery stenosis from X-ray coronary angiography images using deep learning algorithms [7].

More recent research has also explored the potential of AI-assisted clinical decision support systems for coronary angiography interpretation. Studies evaluating AI diagnostic performance compared with cardiologists have shown promising results, indicating that AI tools can support clinicians in interpreting angiographic images and identifying coronary artery abnormalities [8]. Other studies have proposed non-invasive deep learning approaches for estimating fractional flow reserve (FFR), which is an important indicator of coronary artery disease severity [9].

In addition, multimodal AI models have been developed to analyse both imaging and clinical data to improve coronary artery classification and diagnostic accuracy in angiography systems [10]. Research evaluating quantitative flow ratio (QFR) and CT-derived fractional flow reserve (CT-FFR) has also shown improved diagnostic performance for identifying ischemia-causing coronary lesions [11]. Furthermore, novel deep learning approaches have been introduced to detect structural abnormalities such as coronary artery tortuosity in angiographic images, further expanding the capabilities of AI in cardiovascular diagnostics [12].

Despite these advancements, many existing angiogram analysis systems still have several important limitations. Many models focus primarily on detecting the presence of coronary artery disease without identifying the exact artery segment affected or estimating the precise blockage percentage. In addition, several AI models lack explainability and provide limited visual interpretation of detected abnormalities, making it difficult for clinicians to understand the reasoning behind predictions.

Another limitation is the lack of patient-friendly visualization of coronary artery conditions. Traditional angiography images are complex and difficult for non-specialists to interpret, and many existing systems do not generate simplified visual explanations that help patients understand their condition. This reduces the practical usefulness of AI systems in real clinical environments.

Furthermore, most existing angiogram analysis systems do not integrate automated treatment recommendations. Although many models can detect coronary blockages, they do not provide guidance regarding suitable treatments such as medical therapy, percutaneous coronary intervention (PCI), or coronary artery bypass graft surgery (CABG). As a result, clinicians must still manually interpret diagnostic outputs and determine treatment strategies.

To address these limitations, the proposed research introduces an AI-based angiogram evaluation and treatment recommendation component that integrates coronary artery segmentation, blockage detection, stenosis quantification, visual explanation generation, and treatment recommendation within a single intelligent system. By combining computer vision, deep learning models, and visualization technologies, the proposed approach aims to support cardiologists in analysing angiogram data while generating clear visual explanations and evidence-based treatment recommendations.

1.3. Research Gap

Although many studies have explored the use of artificial intelligence for detecting coronary artery disease from medical images, several limitations still exist in current angiogram analysis systems. Many deep learning models focus mainly on classifying angiogram images as normal or abnormal without providing detailed clinical information such as the affected coronary artery, the exact location of stenosis, or the percentage of arterial blockage. However, in real clinical practice, cardiologists require more detailed analysis to determine appropriate treatment strategies. Therefore, the lack of artery-level identification and accurate stenosis quantification represents a significant research gap in existing angiogram-based diagnostic systems.

Another limitation of current research is the lack of explainability in AI-based angiogram analysis models. Many systems operate as “black box” classifiers that provide prediction results without clearly indicating the image regions responsible for the decision. This reduces trust in AI-assisted diagnostic systems and makes it difficult for clinicians to validate the results. Furthermore, traditional angiogram images are complex and difficult for non-specialists to interpret. Existing studies rarely focus on generating clear visual explanations that highlight blockage regions and simplify coronary artery conditions for better understanding.

In addition, most current angiogram analysis approaches focus only on disease detection and do not include treatment recommendation. In clinical cardiology, treatment decisions depend on the severity and distribution of coronary artery blockages. Mild stenosis may require medication and lifestyle management, while significant blockage may require percutaneous coronary intervention (PCI) or coronary artery bypass graft surgery (CABG). However, many AI-based angiogram systems do not integrate treatment decision support based on clinical guidelines, creating another important research gap.

Furthermore, existing studies often focus either on image analysis or clinical decision support separately, rather than integrating both capabilities into a single system. As a result, clinicians must still manually interpret angiogram findings and determine treatment strategies, which may increase workload and introduce variability in decision-making.

To address these limitations, this research proposes an AI-assisted angiogram evaluation and treatment recommendation component that integrates coronary artery segmentation, blockage detection, stenosis quantification, explainable visual outputs, and treatment recommendation within a unified system. The proposed approach aims to identify the affected coronary artery, estimate blockage percentage, classify disease severity, highlight blockage regions, and recommend appropriate treatment options based on clinical guidelines.

By addressing these gaps, the proposed component aims to develop a more interpretable and clinically useful AI-assisted diagnostic system. The integration of automated angiogram analysis with visual explanation and treatment recommendation can support cardiologists in clinical decision-making while improving patient understanding of coronary artery disease conditions.

Feature / Capability	Traditional Angiogram Interpretation	AI Disease Detection Models	Segmentation-Based Research Systems	Clinical Decision Support Tools	Proposed AI Angiogram Evaluation & Treatment Component
Automated Artery Identification	No	Limited	Partial	No	Yes
Stenosis Percentage Estimation	Manual	Limited	Yes	Limited	Yes
Blockage Location Detection	Manual	Partial	Yes	Limited	Yes
Explainable Visual Highlighting	No	Limited	Partial	Limited	Yes
3D Coronary Visualization	No	No	No	No	Yes
Treatment Recommendation Support	No	No	No	Partial	Yes
Multi-Vessel Disease Detection	Manual	Limited	Partial	Limited	Yes
Patient-Friendly Visualization	No	No	Limited	Limited	Yes
Integrated Image Analysis + Decision Support	No	No	No	Partial	Yes

Table 3: Comparative Analysis of Existing Angiogram Analysis Systems

2. OBJECTIVE

2.1. Main Objective

To develop an AI-assisted angiogram evaluation and treatment recommendation component that automatically analyzes coronary angiogram images and reports to detect coronary artery blockages, estimate stenosis severity, and provide explainable visual outputs with clinically relevant treatment recommendations.

2.2. Specific Objectives

- Develop an automated coronary artery segmentation mechanism to identify major coronary arteries such as the Left Anterior Descending (LAD), Right Coronary Artery (RCA), and Left Circumflex (LCX) from coronary angiogram images using deep learning-based image processing techniques.
- Design a deep learning-based model to detect coronary artery blockages and estimate the percentage of arterial stenosis from angiogram images to support accurate diagnosis of coronary artery disease.
- Classify the severity of coronary artery disease by analysing the estimated stenosis percentage and categorising the condition into mild, moderate, or severe blockage levels.
- Generate explainable visual outputs by highlighting detected blockage regions in angiogram images and providing simplified visual representations to improve interpretability for clinicians and patients.
- Develop an AI-based treatment recommendation mechanism that suggests suitable treatment options such as medical management, percutaneous coronary intervention (PCI), or coronary artery bypass graft surgery (CABG) based on blockage severity.
- Evaluate the performance of the proposed component using appropriate evaluation metrics such as accuracy, precision, recall, and segmentation performance for blockage detection and severity classification.

3. METHODOLOGY

The proposed research aims to develop an AI-assisted angiogram evaluation and treatment recommendation component capable of automatically detecting coronary artery blockages, estimating stenosis severity, and recommending appropriate treatment options. The methodology integrates deep learning-based image analysis, medical data processing, and explainable visualization techniques to support clinical decision-making.

3.1 Research Approach

This study adopts a Design Science Research (DSR) approach, which focuses on designing and evaluating an innovative technological artifact to address a real-world problem. In this research, the artifact is an AI-based system capable of analysing coronary angiogram images and generating clinical insights.

The design science approach is appropriate because the research aims to create a functional system that integrates image processing, deep learning, and treatment recommendation mechanisms to support cardiology diagnostics. The development process will involve problem identification, system design, model implementation, evaluation, and iterative refinement.

Additionally, an experimental evaluation strategy will be used to assess the performance of the developed models using medical imaging datasets. The system will be tested to measure its ability to detect coronary artery blockages, estimate stenosis percentage, and classify disease severity accurately.

3.2 Technical Methods and Algorithms

The proposed component will use several artificial intelligence and computer vision techniques to analyse coronary angiogram images and generate diagnostic insights.

Image Preprocessing

Before analysis, angiogram images will undergo preprocessing steps to improve image quality and ensure consistent input for the AI models. These steps include:

- Image resizing and normalization
- Noise reduction and contrast enhancement
- Conversion to grayscale (if required)
- Removal of irrelevant image regions

These preprocessing techniques help improve model performance by emphasizing relevant vascular structures.

Coronary Artery Segmentation

To accurately identify coronary arteries, a U-Net-based deep learning segmentation model will be used to extract vascular structures from angiogram images. Segmentation allows the system to isolate coronary arteries and identifies major branches such as:

- Left Anterior Descending (LAD)
- Right Coronary Artery (RCA)
- Left Circumflex (LCX)
- Left Main Coronary Artery (LMCA)

Segmentation helps improve the accuracy of blockage detection by focusing analysis on relevant vessel regions.

Blockage Detection and Stenosis Estimation

A Convolutional Neural Network (CNN) will be used to analyse the segmented angiogram images to detect narrowing regions in coronary arteries. The model will extract visual features and identify patterns associated with arterial stenosis.

Once a narrowing region is detected, image analysis techniques will estimate the percentage of stenosis by comparing vessel diameter at the blockage location with the normal vessel diameter.

The stenosis severity will be classified as:

- Mild: less than 40% blockage
- Moderate: 40% – 70% blockage

- Severe: greater than 70% blockage

This classification supports clinical decision-making for treatment planning.

Multi-Vessel Disease Identification

The system will also analyse whether the patient has single-vessel, double-vessel, or triple-vessel disease by detecting blockages across multiple coronary arteries. This information is critical for determining suitable treatment strategies.

Explainable Visualization

To improve interpretability, the system will generate visual explanations by highlighting blockage regions on angiogram images. Detected blockage areas will be marked with color indicators and labelled with blockage percentage and artery name.

Additionally, a simplified 3D visualization of coronary arteries will be generated to illustrate the severity and location of blockages. In this visualization:

- Healthy arteries will appear in green
- Narrowed arteries will appear in yellow
- Severe blockages will appear in red

This approach improves understanding for both clinicians and patients.

Treatment Recommendation Mechanism

Based on blockage severity and affected artery location, the system will generate treatment recommendations using rule-based clinical decision logic aligned with cardiology guidelines.

Possible treatment recommendations include:

- Medical Management – recommended for mild stenosis
- Percutaneous Coronary Intervention (PCI) – recommended for significant localised blockage
- Coronary Artery Bypass Graft Surgery (CABG) – recommended for severe multi-vessel disease

This mechanism integrates diagnostic results with treatment guidance.

3.3 Tools and Platforms with Justification

Several software tools and frameworks will be used to develop the proposed system.

Tool / Platform	Purpose
Python	Main programming language for AI development
TensorFlow / PyTorch	Deep learning model development
OpenCV	Image processing and feature extraction
Scikit-learn	Data processing and evaluation

NumPy & Pandas	Data manipulation and preprocessing
Three.js / VTK	3D visualization of coronary arteries
REST APIs	Integration of system modules
Web-based interface	Visualization and interaction with outputs

Table 4 : 3.3 Tools and Platforms with Justification

Python is selected due to its extensive ecosystem for machine learning and medical image analysis. TensorFlow and PyTorch provide powerful deep learning frameworks for building CNN-based models, while OpenCV supports efficient image processing operations.

3.4 Data Collection and Ethical Considerations

The system will be trained and evaluated using publicly available medical imaging datasets and anonymised coronary angiogram images from research repositories. Example datasets include:

- Public cardiovascular imaging datasets
- Medical research image repositories
- Synthetic or augmented angiogram images for model training

All datasets used will contain anonymised patient data without personally identifiable information such as patient name, address, or identification numbers.

Ethical considerations will include:

- Use of anonymised and publicly available datasets
- Protection of patient privacy and confidentiality
- Compliance with medical data usage regulations
- Responsible use of AI in healthcare decision support

The system is designed as a clinical decision support tool and will not replace professional medical judgment.

3.5 Validation Strategy and Performance Metrics

The proposed system will be evaluated using quantitative performance metrics commonly used in medical image analysis.

Model Evaluation Metrics

For blockage detection and classification:

- Accuracy
- Precision

- Recall
- F1-score

For artery segmentation performance:

- Intersection over Union (IoU)
- Dice Similarity Coefficient (DSC)

For stenosis estimation:

- Mean Absolute Error (MAE) between predicted and actual stenosis percentage.

Validation Strategy

The dataset will be divided into:

- Training dataset
- Validation dataset
- Testing dataset

Cross-validation techniques will be applied to ensure robust evaluation of model performance.

Additionally, the visual outputs and treatment recommendations generated by the system will be reviewed to ensure they align with clinical cardiology guidelines.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of Proposed Solution and Its Components

The proposed solution is an AI-assisted Angiogram Evaluation and Treatment Recommendation component that analyses coronary angiogram images and reports to identify arterial blockages, estimate stenosis severity, and recommend suitable treatment options. According to the project workflow, this component is activated in the cardiac pathway after earlier screening and validation stages, and its role is to measure stenosis percentage, identify affected arteries, and support treatment decisions such as medical management, PCI, or CABG. The component is designed as an intelligent clinical decision support module that combines image preprocessing, artery segmentation, blockage detection, severity estimation, visualisation, and treatment recommendation within a single workflow.

The architecture consists of three major layers: the Input Layer, Processing Layer, and Output Layer. The Input Layer receives angiogram images and related cardiac assessment inputs. The Processing Layer performs preprocessing, AI-based blockage detection, severity analysis, and treatment generation. The Output Layer delivers blockage severity, treatment recommendations, predicted outcomes, and follow-up guidance in a structured and understandable format.

4.2 System Diagram of Components

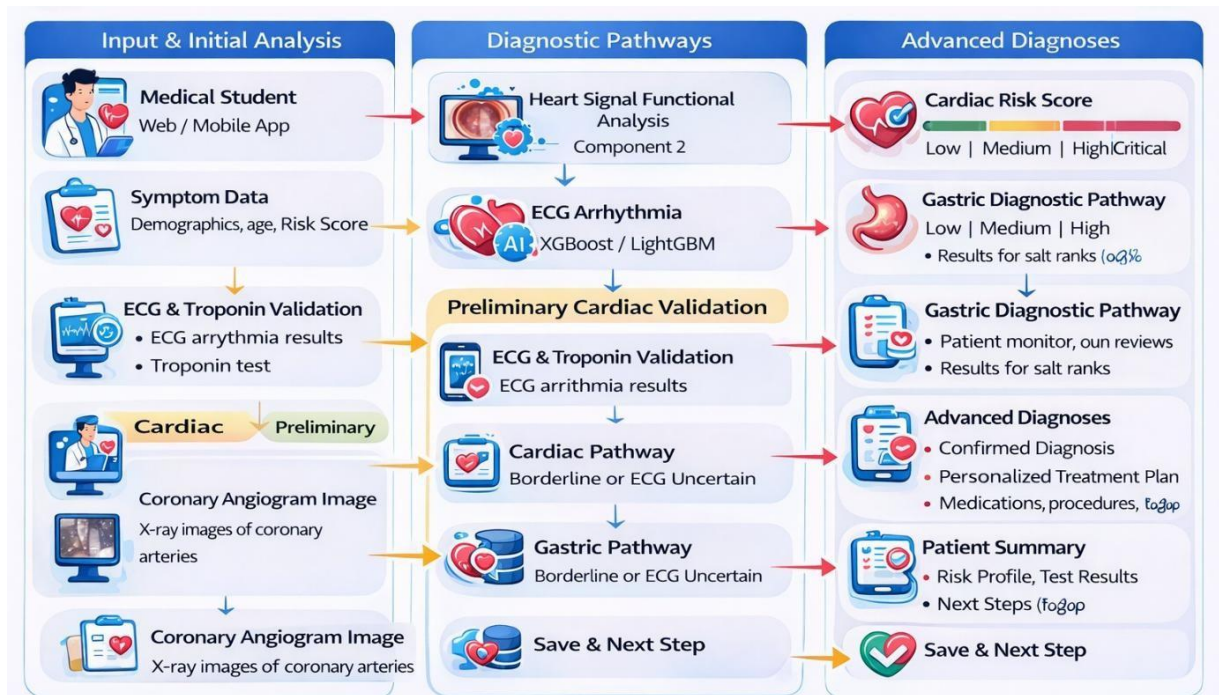


Figure 1 : 4.2 System Diagram of Components

4.3 Individual Component and Its Parts

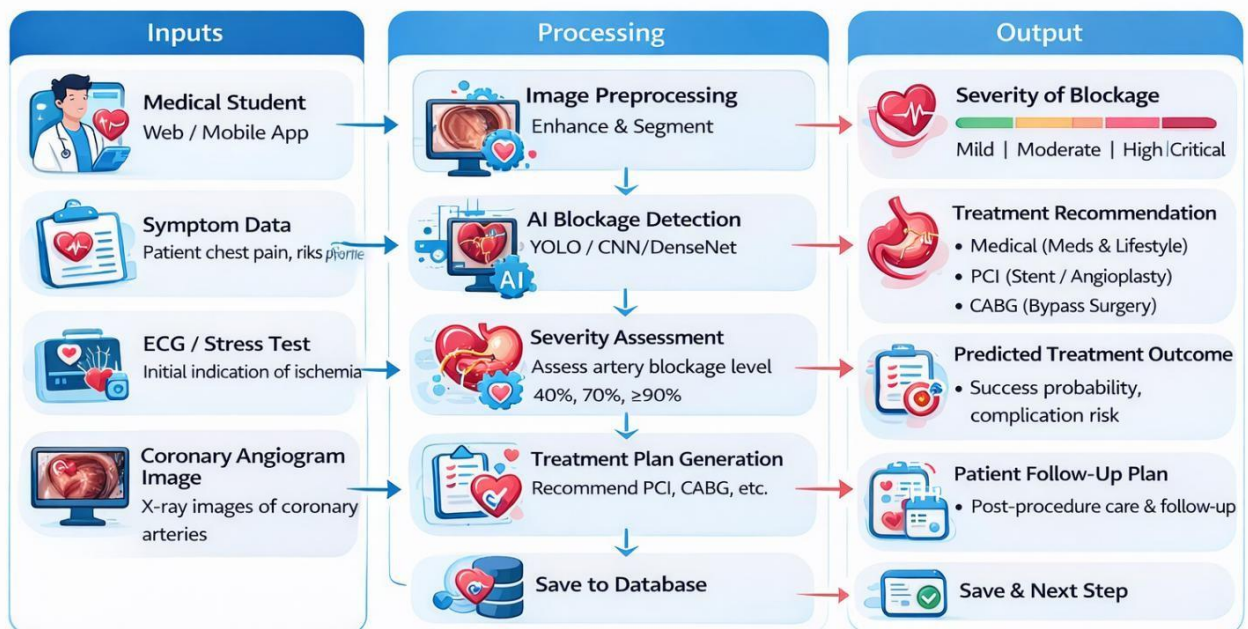


Figure 2: 4.3 Individual Component and Its Parts

The Input Layer contains the data sources required by the component. These include coronary angiogram images as the primary imaging input, and related supporting information such as symptom data, ECG/stress test findings, and clinician or medical student interaction through a web or mobile

interface. In the broader project workflow, angiogram analysis occurs after cardiac confirmation, making coronary angiography the central input for this component.

The Processing Layer is the core analytical section of the architecture. It begins with image preprocessing, where angiogram images are resized, enhanced, normalized, and segmented to isolate relevant artery structures. After preprocessing, the system performs AI blockage detection using deep learning techniques such as CNN-based models and artery segmentation methods. The next stage is severity assessment, where the system estimates blockage level and classifies stenosis severity. Based on these findings, a treatment plan generation module recommends suitable interventions such as medical therapy, PCI, or CABG. Finally, the processed results are stored for future retrieval and analysis. These stages are directly consistent with the methodology and expected outputs defined for Component 3.

The Output Layer presents the final system results. These include the severity of blockage, treatment recommendation, predicted treatment outcome, patient follow-up plan, and save/next-step actions. This output structure supports both clinicians and non-medical users by translating technical angiogram findings into understandable clinical recommendations. The component document also emphasizes visual explanation and patient-friendly understanding as key contributions of this module.

4.4 Component Interactions

The component interacts internally through a sequential workflow. First, the input data is transferred into the preprocessing module, where the angiogram image is prepared for analysis. The preprocessed image is then passed to the AI blockage detection model, which identifies affected coronary arteries and narrowing regions. The detected results are forwarded to the severity assessment module, which calculates stenosis percentage and classifies disease severity. Based on this classification, the treatment recommendation engine generates clinically relevant intervention suggestions.

There are also external interactions with the wider project pipeline. Earlier modules in the overall workflow provide prior validation through ECG, troponin, and cardiac pathway selection before the patient reaches angiogram evaluation. After Component 3 completes its analysis, the outputs can be used by clinicians for treatment planning and case review. This interaction matches the project workflow in which angiogram analysis is positioned after cardiac functional testing and before treatment pathway selection.

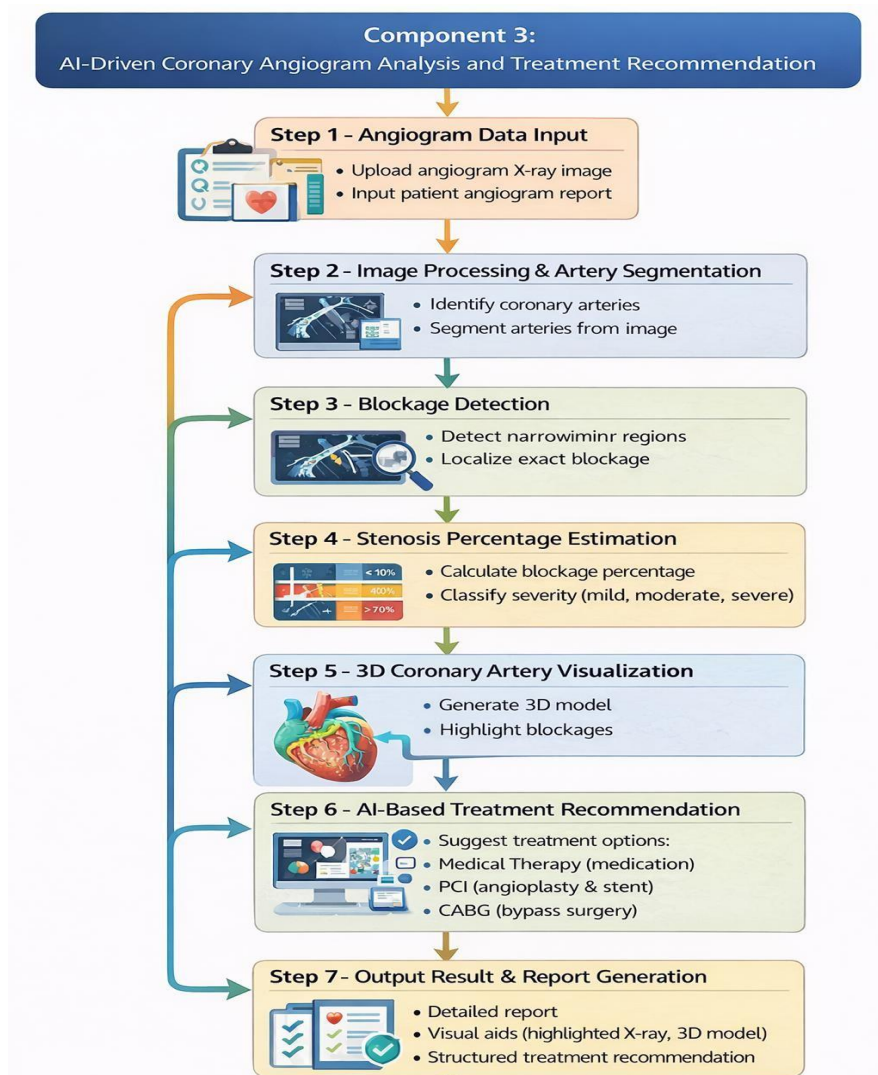


Figure 3: 4.4 Component Interactions

4.5 Data Flow Explanation

The data flow of this component starts when a coronary angiogram image and related clinical details are uploaded into the system. The image first moves into the preprocessing stage, where quality enhancement and artery segmentation are performed. The processed image then enters the AI model, which detects blockage regions and identifies the affected artery. The severity assessment stage calculates stenosis level and categorizes the condition into severity classes. Finally, these analytical results are passed to the treatment recommendation module, which produces suitable treatment options and explanatory outputs.

This data flow ensures that raw angiogram data is transformed into clinically meaningful information step by step. It also supports the expected outputs described for the component, including affected artery identification, blockage percentage, severity classification, enhanced angiogram visualization, and treatment recommendation.

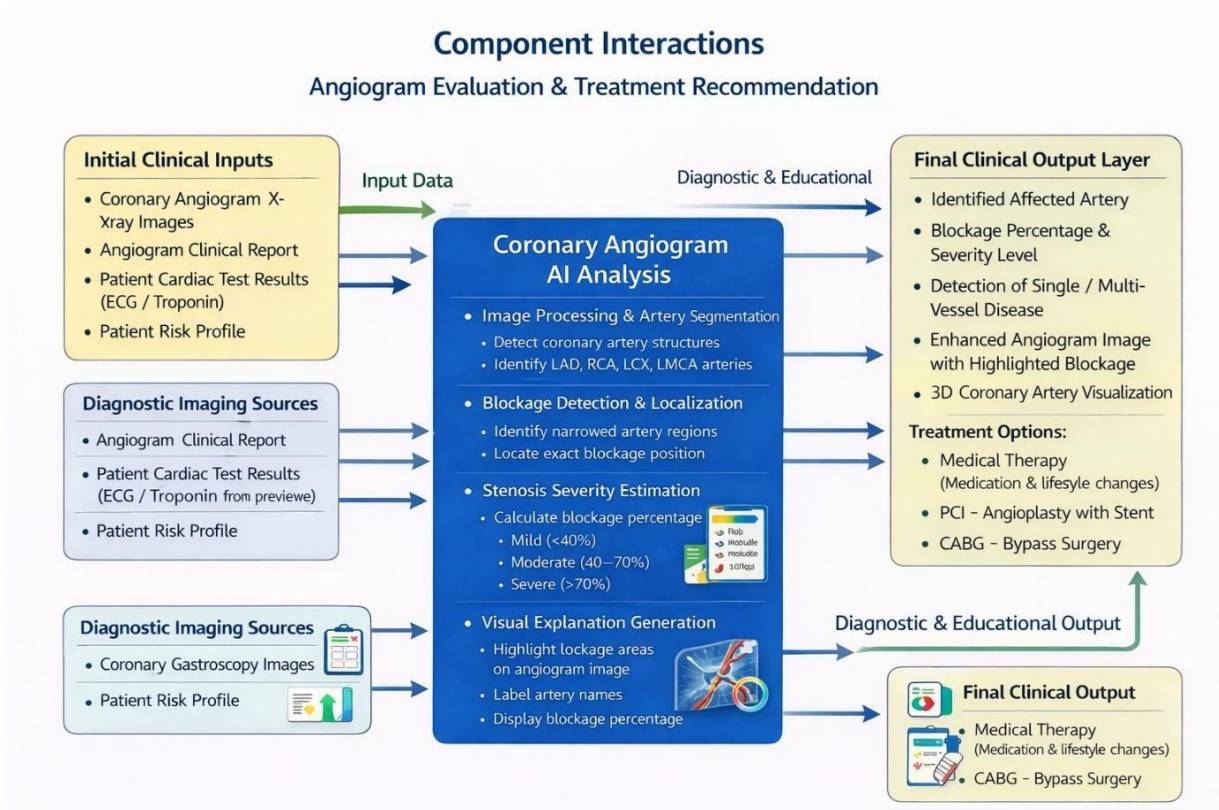


Figure 4: 4.5 Data Flow Explanation

4.6 Scalability and Security Considerations

The proposed architecture is designed to be scalable so that it can handle increasing volumes of angiogram images and future feature expansion. Scalability can be achieved by deploying the AI models and storage components on cloud-based infrastructure, which allows larger datasets, more users, and faster model processing when the system is adopted in clinical environments. The modular design of the architecture also makes it easier to update individual components, such as the blockage detection model or visualization module, without redesigning the entire system.

Security is a critical requirement because the component handles sensitive medical imaging data. All uploaded angiogram images and related clinical information should be stored securely using access control, encrypted storage, and authenticated user access. Patient-identifiable information should be minimized or anonymized wherever possible, and only authorized medical users should be allowed to view diagnostic outputs. These security measures are necessary to protect privacy and ensure safe use of the system in healthcare settings.

5. USER REQUIREMENTS

5.1. Stakeholder Identification

Several stakeholders will interact with or benefit from the proposed system.

Cardiologists

Cardiologists are the primary users of the system. They will use the component to analyse coronary angiogram images, identify arterial blockages, and review AI-generated treatment recommendations to support clinical decision-making.

Medical Students and Trainee Doctors

Medical students and trainee doctors can use the system as an educational tool to understand angiogram interpretation, coronary artery anatomy, and treatment selection for coronary artery disease.

Patients

Patients benefit indirectly from the system through improved diagnostic explanations and visualisations of coronary artery conditions. Simplified visual outputs help patients better understand their medical condition.

Healthcare Institutions

Hospitals and medical centres may use the system to improve diagnostic efficiency, reduce interpretation time, and support clinical decision-making using AI-assisted analysis.

Researchers and Developers

Researchers and system developers may use the platform to further improve AI models for medical imaging analysis and clinical decision support.

5.2. Functional Requirements

The system shall automatically identify and segment major coronary arteries such as the Left Anterior Descending (LAD), Right Coronary Artery (RCA), and Left Circumflex (LCX).

FR3 – Blockage Detection

The system shall detect narrowing regions in coronary arteries from angiogram images.

FR4 – Stenosis Percentage Estimation

The system shall estimate the percentage of arterial blockage based on vessel diameter analysis.

FR5 – Disease Severity Classification

The system shall classify coronary artery disease severity as mild, moderate, or severe based on stenosis percentage.

FR6 – Multi-Vessel Disease Identification

The system shall determine whether the patient has single-vessel, double-vessel, or triple-vessel disease.

FR7 – Visual Highlight Generation

The system shall generate visual outputs that highlight blockage areas within angiogram images.

FR8 – 3D Coronary Artery Visualisation

The system shall generate simplified 3D visual representations of coronary arteries showing healthy and blocked regions.

FR9 – Treatment Recommendation

The system shall recommend appropriate treatment options such as medical therapy, percutaneous coronary intervention (PCI), or coronary artery bypass graft surgery (CABG) based on blockage severity.

FR10 – Result Display

The system shall present analysis results, including affected artery, blockage percentage, severity classification, and treatment recommendation.

5.3 Non-Functional Requirements

Non-functional requirements describe system quality attributes such as performance, security, reliability, and usability. These requirements ensure that the system operates efficiently and safely within a healthcare environment.

Performance

The system should process angiogram images and generate analysis results within a reasonable time. This is a non-functional requirement because it defines how efficiently the system performs its tasks rather than what tasks it performs.

Accuracy

The AI model should achieve high diagnostic accuracy in detecting blockages and estimating stenosis severity.

This is a non-functional requirement because it defines the quality and reliability of system outputs rather than a specific function.

Security

The system must protect sensitive medical data and ensure that uploaded medical images are securely handled.

This is a non-functional requirement because it relates to data protection and system safety rather than system functionality.

Usability

The system interface should be simple and easy to use for clinicians and medical students.

This is a non-functional requirement because it focuses on user experience and ease of interaction rather than system capabilities.

Reliability

The system should consistently produce stable results and operate without failure during analysis. This is a non-functional requirement because it describes system dependability rather than a specific function.

Scalability

The system should be capable of handling increasing numbers of angiogram images and users without performance degradation.

This is a non-functional requirement because it describes system capacity and growth capability rather than functional behaviour.

Interpretability

The system should provide visual explanations that help clinicians understand AI-generated predictions.

This is a non-functional requirement because it improves transparency and trust in the system rather than introducing a new function.

6. COMMERCIALIZATION PLAN

The proposed AI-assisted angiogram evaluation and treatment recommendation component has potential for commercialization as a clinical decision support tool for cardiology diagnostics. By integrating artificial intelligence with coronary angiogram analysis, the system aims to improve diagnostic efficiency, assist clinicians in treatment planning, and enhance patient understanding of coronary artery disease. The commercialization strategy focuses on deploying the system within healthcare institutions and medical training environments.

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes hospitals, cardiology clinics, and diagnostic imaging centres that perform coronary angiography procedures. These healthcare institutions require efficient tools to assist cardiologists in interpreting angiogram images and making informed treatment decisions.

Another potential market includes medical universities and training institutions, where the system can be used as an educational platform for teaching angiogram interpretation and coronary artery disease management.

The key customer personas include:

Cardiologists and Interventional Cardiologists – Professionals responsible for diagnosing and treating coronary artery disease who require accurate and efficient analysis of angiogram images.

Hospital Administrators – Decision-makers responsible for adopting technologies that improve clinical efficiency and patient care.

Medical Educators and Students – Individuals who can use the system as a learning tool to understand coronary artery anatomy and treatment strategies.

6.2 Value Proposition

The proposed system offers several key benefits to healthcare providers and patients.

First, it improves diagnostic efficiency by automatically detecting coronary artery blockages and estimating stenosis severity from angiogram images. This can reduce the time required for manual interpretation and support faster clinical decision-making.

Second, the system supports treatment planning by recommending appropriate interventions such as medication, percutaneous coronary intervention (PCI), or coronary artery bypass graft surgery (CABG) based on detected blockage severity.

Third, the system enhances interpretability by generating visual explanations and simplified 3D visualizations of coronary arteries. These visual outputs help clinicians understand AI predictions and allow patients to better understand their medical condition.

Overall, the proposed solution aims to improve the accuracy, efficiency, and transparency of coronary artery disease diagnosis.

6.3 Revenue Model

The proposed system can be commercialized using several revenue models.

One potential approach is a Software-as-a-Service (SaaS) model, where hospitals and healthcare institutions subscribe to the platform and pay a monthly or annual licensing fee for using the AI-based diagnostic system.

Another option is a per-analysis pricing model, where healthcare institutions pay a small fee for each angiogram image analysed by the system.

The system can also be offered as an enterprise software solution integrated into existing hospital information systems or medical imaging platforms.

Additionally, the platform may be licensed to medical universities and training centres as an educational tool for cardiology training.

6.4 Cost Estimation

The cost of developing and maintaining the system includes several components:

Development Costs

- AI model development and training
- Software development and system integration
- User interface development

Infrastructure Costs

- Cloud computing services for AI model training and deployment
- Data storage for angiogram images and analysis results

Operational Costs

- System maintenance and updates
- Technical support for healthcare institutions

Research and Testing Costs

- Dataset acquisition and preprocessing
- Model evaluation and validation

Using open-source technologies such as Python, TensorFlow, and OpenCV helps reduce development costs.

6.5 Pricing Strategy

The pricing strategy should ensure affordability for healthcare institutions while maintaining sustainability for system maintenance.

A tiered subscription model can be implemented:

- Basic Plan – AI-based angiogram analysis and blockage detection
- Advanced Plan – Includes visual explanations and treatment recommendations
- Enterprise Plan – Full system integration with hospital information systems and imaging platforms

This flexible pricing structure allows institutions of different sizes to adopt the technology based on their requirements.

6.6 Competitive Advantage

The proposed system provides several advantages compared with existing solutions.

First, it integrates angiogram analysis, stenosis quantification, visual explanation, and treatment recommendation within a single platform, whereas many existing systems focus only on disease detection.

Second, the system generates patient-friendly visualisations, including highlighted angiogram images and simplified 3D coronary artery models, which improve interpretability for both clinicians and patients.

Third, the system combines AI-driven image analysis with clinical decision support, helping cardiologists select appropriate treatment strategies based on blockage severity.

These capabilities create a more comprehensive solution for coronary artery disease analysis compared with existing diagnostic tools.

6.7 Intellectual Property Considerations

To protect the innovation and commercialisation potential of the proposed system, several intellectual property strategies may be considered.

The AI algorithms and system architecture can be protected through software copyright and intellectual property registration. Additionally, unique methods used for angiogram analysis, visualisation, or treatment recommendation may be eligible for patent protection depending on their novelty.

Furthermore, licensing agreements may be established for using publicly available medical datasets while ensuring compliance with data usage regulations.

Proper intellectual property protection ensures that the developed technology can be commercialised while protecting the rights of the developers and research team.

7. BUDGET AND JUSTIFICATION

The budget for Component 3 focuses on developing the AI-assisted Angiogram Evaluation and Treatment Recommendation system, which analyzes angiogram images, detects coronary artery blockages, estimates stenosis percentage, generates visual explanations, and recommends treatment options. The development requires computing resources, data storage, cloud infrastructure, and research support. The total estimated budget for this component is LKR 78,000, which falls within the expected range for AI-based research projects.

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Purpose
External Hard Drive (1TB)	18,000	Storage for angiogram image datasets, trained AI models, and experimental outputs.
GPU Upgrade / High-performance workstation access	15,000	Required for training deep learning models such as CNN and U-Net used for artery segmentation and blockage detection.
Backup Storage (USB / SSD)	5,000	Backup of datasets, trained models, and experimental results to avoid data loss.
Total Hardware Cost	38,000 LKR	

Table 5 : 7.1 Hardware Costs

Justification

Deep learning models used in angiogram image analysis require high computational resources and large data storage. Since angiogram images and trained models consume significant space, dedicated storage devices are necessary. GPU processing support is also required for efficient training of AI models.

7.2 Target Market and Customer Persona

Item	Estimated Cost (LKR)	Purpose
Visualisation Tools (Blender / Unity plugins)	4,000	Used for generating simplified 3D coronary artery visualisations showing blockage severity.
Development Tools and IDE Support	2,000	Tools used for system interface development and debugging.
Total Software Cost	6,000 LKR	

Table 6 : 7.2 Target Market and Customer Persona

Justification

Although most AI development tools such as Python, TensorFlow, PyTorch, and OpenCV are open-source, additional visualization or development utilities may be required for implementing 3D coronary artery visualization, which is a key feature of the proposed system.

7.3 Cloud Services

Item	Estimated Cost (LKR)	Purpose
Cloud GPU Computing (Google Cloud / AWS / Colab Pro)	25,000	Training deep learning models for angiogram segmentation, blockage detection, and stenosis estimation.
Cloud Data Storage	5,000	Secure storage of angiogram datasets and trained models for remote access and collaboration.
Total Cloud Cost	30,000 LKR	

Table 7:7.3 Cloud service

Justification

Training convolutional neural networks and segmentation models for angiogram analysis requires high GPU processing power. Cloud platforms provide scalable computing resources that allow faster model training and experimentation compared to standard personal computers.

7.4 Human Effort Cost

Item	Estimated Cost (LKR)	Purpose
Internet and Research Resources	3,000	Accessing medical datasets, downloading research papers, and collaborating with team members.
Documentation and Report Preparation	1,000	Printing, diagrams, and documentation preparation for proposal and project reports.

Total Human Effort Cost:	4,000 LKR	
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Table 8:7.4 Human Effort Cost

Justification

Research activities require access to online datasets, research publications, and collaboration platforms. Additionally, documentation costs are necessary for preparing research reports, diagrams, and evaluation materials required for the final project submission.

7.5 Total Estimated Budget

Category	Cost (LKR)
Hardware Costs	38,000
Software Licenses	6,000
Cloud Services	30,000
Human Effort Cost	4,000
Total Estimated Budget	78,000 LKR

Table 9 : 7.6 Total Estimated Budget

Justification

The total estimated budget of LKR 78,000 is allocated to support the development of the AI-assisted Angiogram Evaluation and Treatment Recommendation component. Most of the cost is dedicated to hardware and cloud computing resources required for training deep learning models and storing angiogram datasets. Additional expenses cover software tools, internet access, and documentation needed for system development and research activities. Using open-source technologies helps reduce overall costs while ensuring sufficient resources for successful implementation.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown

Task ID	Task Name	Description	Timeline
T1	Dataset Collection	Collect publicly available coronary angiogram images and reports from medical datasets and research repositories.	March – April 2026
T2	Data Preparation	Annotate angiogram images, label affected arteries, and prepare the dataset for training AI models.	April 2026
T3	Image Preprocessing	Perform image resizing, normalization, noise reduction, and dataset splitting into training and testing sets.	April – May 2026
T4	Artery Segmentation Model	Develop a U-Net based model to identify and segment coronary arteries from angiogram images.	May – June 2026
T5	Blockage Detection Model	Train CNN-based models to detect narrowing regions and identify blocked arteries.	June – July 2026
T6	Stenosis Severity Estimation	Calculate blockage percentage and classify disease severity into mild, moderate, and severe levels.	July 2026
T7	Visualization Module	Generate highlighted angiogram images and simplified 3D coronary artery visualization.	August 2026
T8	Treatment Recommendation System	Implement rule-based AI logic to recommend treatments such as medication, PCI, or CABG.	August – September 2026
T9	System Integration	Integrate all modules into a unified system with a user interface.	September 2026
T10	Testing and Validation	Evaluate system performance using accuracy, precision, recall, and segmentation metrics.	September – October 2026

Table 10 : 8.1 Task Breakdown

8.2 Timeline Alignment

The tasks for the Angiogram Evaluation and Treatment Recommendation component are aligned with the official project timeline to ensure timely completion. Early stages focus on dataset preparation and model development, while later stages focus on system integration, testing, and validation before the final report and presentation deadlines.

Phase	Task Activities	Timeline	Related RP Milestone
Phase 1	Dataset collection and data preparation	March – April 2026	Proposal Submission & Presentation
Phase 2	Image preprocessing and dataset splitting	April – May 2026	Progress Presentation 1
Phase 3	Artery segmentation and blockage detection model development	June – July 2026	Mid project development stage
Phase 4	Stenosis severity estimation and visualization generation	July – August 2026	Progress Presentation 2
Phase 5	Treatment recommendation system development	August – September 2026	System development completion
Phase 6	System integration, testing, and validation	September – October 2026	Final Report Submission & Final Presentation

Table 11 : 8.2 Timeline Alignment

8.3 Balanced Workload Distribution

The workload for this component is distributed across several development phases to avoid overloading a single stage of the project. Initial tasks focus on dataset preparation and preprocessing, which form the foundation for AI model development. The middle phase involves training and optimizing deep learning models for artery segmentation and blockage detection. The final phase focuses on visualization generation, treatment recommendation implementation, and system testing to ensure reliable performance.

This balanced distribution allows sufficient time for experimentation, model refinement, and validation while ensuring that all system components are completed before the final project submission.

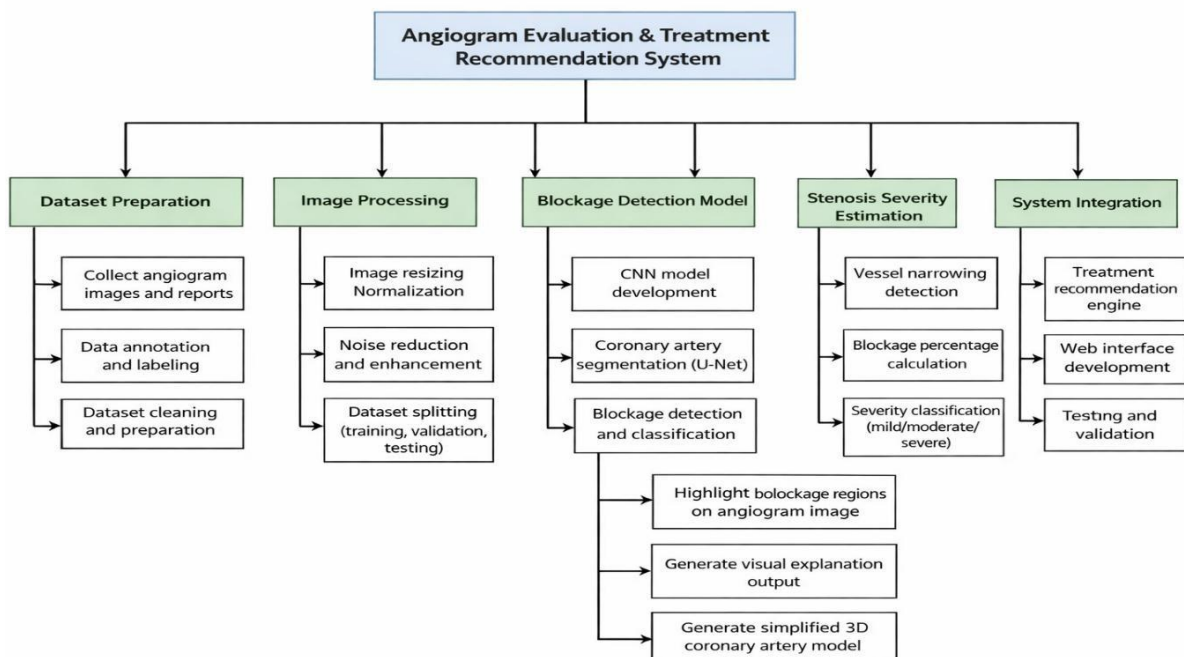


Figure 5:8.3 Balanced Workload Distribution

9. GANTT CHART

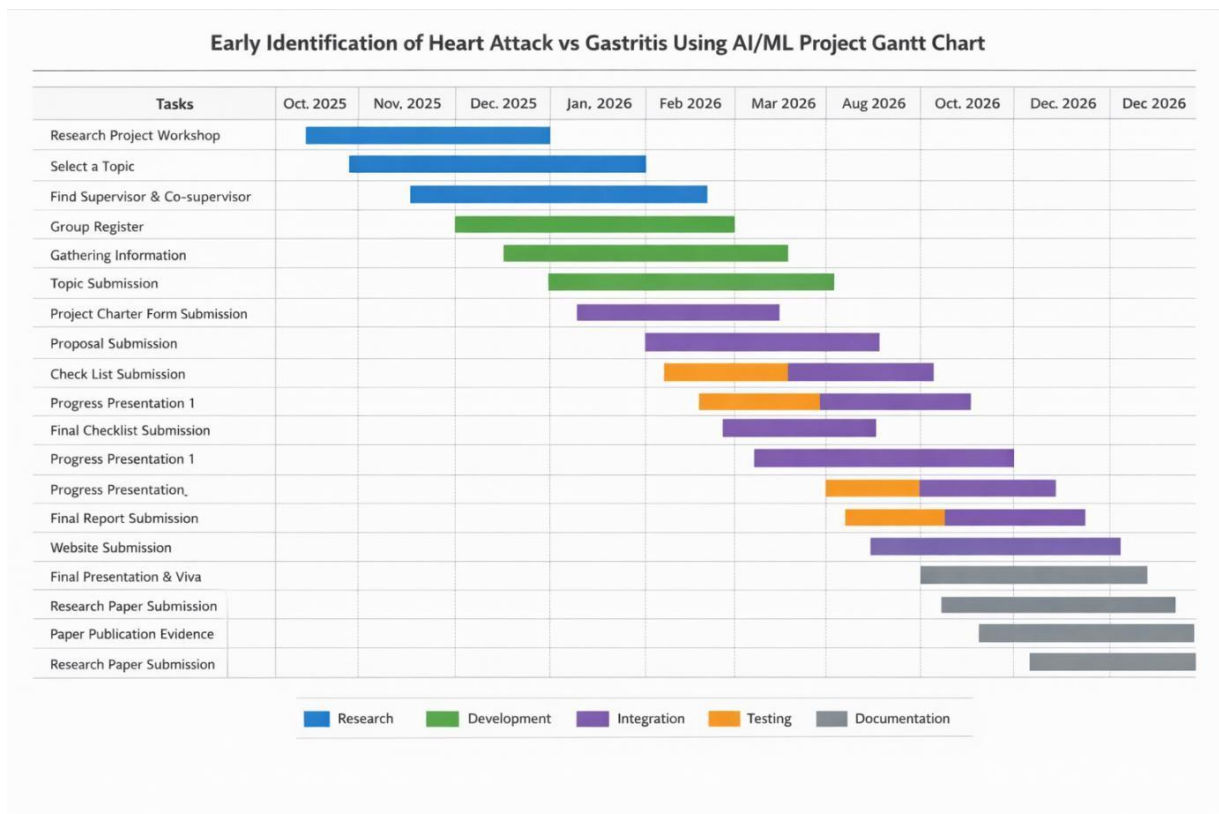


Figure 6: Gantt Chart

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10.APPENDICES

Appendix A – Survey Instruments

This appendix contains the questionnaire used to collect patient symptom data related to chest pain and gastric conditions. The survey includes questions regarding patient demographics, lifestyle factors, meal-related symptoms, medication history, and other relevant clinical indicators that help differentiate cardiac and gastric chest pain.

Appendix B – Preliminary UI Mockups

This appendix includes preliminary user interface designs for the medical student dashboard. The mockups demonstrate how users can upload gastroscopy images, review AI predictions, view severity levels, and access treatment recommendations generated by the system.

Appendix C – Risk Analysis

Potential Risk	Description	Mitigation Strategy
Limited Dataset Availability	Insufficient endoscopy images for training the AI model may affect model performance.	Use publicly available datasets such as HyperKvasir and apply data augmentation techniques.
Model Accuracy Issues	The AI model may produce incorrect predictions due to limited or imbalanced data.	Use validation techniques, improve dataset quality, and tune model parameters.
System Integration Problems	Difficulty integrating the gastric analysis module with other system components.	Use modular architecture and well-defined APIs for communication between components.
Data Privacy Concerns	Handling medical images and patient data may raise privacy issues.	Ensure all datasets are anonymized and implement secure data storage methods.
Hardware or Resource Limitations	Limited computing power may slow down model training and testing.	Use cloud-based computing resources and optimize model training processes.

Table 12: Appendix C – Risk Analysis

Appendix D – Ethical Documents

This appendix includes ethical considerations related to the use of medical datasets and patient information. All patient data used in the research will be anonymised to protect privacy, and datasets will be used strictly for research and educational purposes in accordance with medical data protection guidelines.